

where C is a chain complex of free R -modules with boundary maps R -module homomorphisms, and the coefficient group G is also an R -module. If R is a field, for example, then R -modules are always free and so the Ext_R term is always zero since we may choose free resolutions of the form $0 \rightarrow F_0 \rightarrow H \rightarrow 0$.

It is interesting to note that the proof of Lemma 3.1 on the uniqueness of free resolutions is valid for modules over an arbitrary ring R . Moreover, every R -module H has a free resolution, which can be constructed in the following way. Choose a set of generators for H as an R -module, and let F_0 be a free R -module with basis in one-to-one correspondence with these generators. Thus we have a surjective homomorphism $f_0: F_0 \rightarrow H$ sending the basis elements to the chosen generators. Now repeat the process with $\text{Ker } f_0$ in place of H , constructing a homomorphism $f_1: F_1 \rightarrow F_0$ sending a basis for a free R -module F_1 onto generators for $\text{Ker } f_0$. And inductively, construct $f_n: F_n \rightarrow F_{n-1}$ with image equal to $\text{Ker } f_{n-1}$ by the same procedure.

By Lemma 3.1 the groups $H^n(F; G)$ depend only on H and G , not on the free resolution F . The standard notation for $H^n(F; G)$ is $\text{Ext}_R^n(H, G)$. For sufficiently complicated rings R the groups $\text{Ext}_R^n(H, G)$ can be nonzero for $n > 1$. In certain more advanced topics in algebraic topology these Ext_R^n groups play an essential role.

A final remark about the definition of $\text{Ext}_R^n(H, G)$: By the Exercise stated earlier, exactness of $F_1 \rightarrow F_0 \rightarrow H \rightarrow 0$ implies exactness of $F_1^* \leftarrow F_0^* \leftarrow H^* \leftarrow 0$. This means that $H^0(F; G)$ as defined above is zero. Rather than having $\text{Ext}_R^0(H, G)$ be automatically zero, it is better to define $H^n(F; G)$ as the n^{th} homology group of the complex $\cdots \leftarrow F_1^* \leftarrow F_0^* \leftarrow 0$ with the term H^* omitted. This can be viewed as defining the groups $H^n(F; G)$ to be unreduced cohomology groups. With this slightly modified definition we have $\text{Ext}_R^0(H, G) = H^0(F; G) = H^* = \text{Hom}_R(H, G)$ by the exactness of $F_1^* \leftarrow F_0^* \leftarrow H^* \leftarrow 0$. The real reason why unreduced Ext groups are better than reduced groups is perhaps to be found in certain exact sequences involving Ext and Hom derived in §3.F, which would not work with the Hom terms replaced by zeros.

Cohomology of Spaces

Now we return to topology. Given a space X and an abelian group G , we define the group $C^n(X; G)$ of **singular n -cochains with coefficients in G** to be the dual group $\text{Hom}(C_n(X), G)$ of the singular chain group $C_n(X)$. Thus an n -cochain $\varphi \in C^n(X; G)$ assigns to each singular n -simplex $\sigma: \Delta^n \rightarrow X$ a value $\varphi(\sigma) \in G$. Since the singular n -simplices form a basis for $C_n(X)$, these values can be chosen arbitrarily, hence n -cochains are exactly equivalent to functions from singular n -simplices to G .

The **coboundary map** $\delta: C^n(X; G) \rightarrow C^{n+1}(X; G)$ is the dual ∂^* , so for a cochain $\varphi \in C^n(X; G)$, its coboundary $\delta\varphi$ is the composition $C_{n+1}(X) \xrightarrow{\partial} C_n(X) \xrightarrow{\varphi} G$. This means that for a singular $(n+1)$ -simplex $\sigma: \Delta^{n+1} \rightarrow X$ we have

$$\delta\varphi(\sigma) = \sum_i (-1)^i \varphi(\sigma | [v_0, \dots, \hat{v}_i, \dots, v_{n+1}])$$

It is automatic that $\delta^2 = 0$ since δ^2 is the dual of $\partial^2 = 0$. Therefore we can define the **cohomology group** $H^n(X; G)$ **with coefficients in G** to be the quotient $\text{Ker } \delta / \text{Im } \delta$ at $C^n(X; G)$ in the cochain complex

$$\cdots \leftarrow C^{n+1}(X; G) \xleftarrow{\delta} C^n(X; G) \xleftarrow{\delta} C^{n-1}(X; G) \leftarrow \cdots \leftarrow C^0(X; G) \leftarrow 0$$

Elements of $\text{Ker } \delta$ are **cocycles**, and elements of $\text{Im } \delta$ are **coboundaries**. For a cochain φ to be a cocycle means that $\delta\varphi = \varphi\partial = 0$, or in other words, φ vanishes on boundaries.

Since the chain groups $C_n(X)$ are free, the algebraic universal coefficient theorem takes on the topological guise of split short exact sequences

$$0 \rightarrow \text{Ext}(H_{n-1}(X), G) \rightarrow H^n(X; G) \rightarrow \text{Hom}(H_n(X), G) \rightarrow 0$$

which describe how cohomology groups with arbitrary coefficients are determined purely algebraically by homology groups with $\mathbb{Z}$ coefficients. For example, if the homology groups of X are finitely generated then Corollary 3.3 tells how to compute the cohomology groups $H^n(X; \mathbb{Z})$ from the homology groups.

When $n = 0$ there is no Ext term, and the universal coefficient theorem reduces to an isomorphism $H^0(X; G) \approx \text{Hom}(H_0(X), G)$. This can also be seen directly from the definitions. Since singular 0-simplices are just points of X , a cochain in $C^0(X; G)$ is an arbitrary function $\varphi: X \rightarrow G$, not necessarily continuous. For this to be a cocycle means that for each singular 1-simplex $\sigma: [v_0, v_1] \rightarrow X$ we have $\delta\varphi(\sigma) = \varphi(\partial\sigma) = \varphi(\sigma(v_1)) - \varphi(\sigma(v_0)) = 0$. This is equivalent to saying that φ is constant on path-components of X . Thus $H^0(X; G)$ is all the functions from path-components of X to G . This is the same as $\text{Hom}(H_0(X), G)$.

Likewise in the case of $H^1(X; G)$ the universal coefficient theorem gives an isomorphism $H^1(X; G) \approx \text{Hom}(H_1(X), G)$ since $\text{Ext}(H_0(X), G) = 0$, the group $H_0(X)$ being free. If X is path-connected, $H_1(X)$ is the abelianization of $\pi_1(X)$ and we can identify $\text{Hom}(H_1(X), G)$ with $\text{Hom}(\pi_1(X), G)$ since G is abelian.

The universal coefficient theorem has a simpler form if we take coefficients in a field F for both homology and cohomology. In §2.2 we defined the homology groups $H_n(X; F)$ as the homology groups of the chain complex of free F -modules $C_n(X; F)$, where $C_n(X; F)$ has basis the singular n -simplices in X . The dual complex $\text{Hom}_F(C_n(X; F), F)$ of F -module homomorphisms is the same as $\text{Hom}(C_n(X), F)$ since both can be identified with the functions from singular n -simplices to F . Hence the homology groups of the dual complex $\text{Hom}_F(C_n(X; F), F)$ are the cohomology groups $H^n(X; F)$. In the generalization of the universal coefficient theorem to the case of modules over a principal ideal domain, the Ext_F terms vanish since F is a field, so we obtain isomorphisms

$$H^n(X; F) \approx \text{Hom}_F(H_n(X; F), F)$$

Thus, with field coefficients, cohomology is the exact dual of homology. Note that when $F = \mathbb{Z}_p$ or $\mathbb{Q}$ we have $\text{Hom}_F(H, G) = \text{Hom}(H, G)$, the group homomorphisms, for arbitrary F -modules G and H .

For the remainder of this section we will go through the main features of singular homology and check that they extend without much difficulty to cohomology.

Reduced Groups. Reduced cohomology groups $\tilde{H}^n(X; G)$ can be defined by dualizing the augmented chain complex $\cdots \rightarrow C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0$, then taking Ker / Im . As with homology, this gives $\tilde{H}^n(X; G) = H^n(X; G)$ for $n > 0$, and the universal coefficient theorem identifies $\tilde{H}^0(X; G)$ with $\text{Hom}(\tilde{H}_0(X), G)$. We can describe the difference between $\tilde{H}^0(X; G)$ and $H^0(X; G)$ more explicitly by using the interpretation of $H^0(X; G)$ as functions $X \rightarrow G$ that are constant on path-components. Recall that the augmentation map $\varepsilon: C_0(X) \rightarrow \mathbb{Z}$ sends each singular 0-simplex σ to 1, so the dual map ε^* sends a homomorphism $\varphi: \mathbb{Z} \rightarrow G$ to the composition $C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \xrightarrow{\varphi} G$, which is the function $\sigma \mapsto \varphi(1)$. This is a constant function $X \rightarrow G$, and since $\varphi(1)$ can be any element of G , the image of ε^* consists of precisely the constant functions. Thus $\tilde{H}^0(X; G)$ is all functions $X \rightarrow G$ that are constant on path-components modulo the functions that are constant on all of X .

Relative Groups and the Long Exact Sequence of a Pair. To define relative groups $H^n(X, A; G)$ for a pair (X, A) we first dualize the short exact sequence

$$0 \rightarrow C_n(A) \xrightarrow{i} C_n(X) \xrightarrow{j} C_n(X, A) \rightarrow 0$$

by applying $\text{Hom}(-, G)$ to get

$$0 \leftarrow C^n(A; G) \xleftarrow{i^*} C^n(X; G) \xleftarrow{j^*} C^n(X, A; G) \leftarrow 0$$

where by definition $C^n(X, A; G) = \text{Hom}(C_n(X, A), G)$. This sequence is exact by the following direct argument. The map i^* restricts a cochain on X to a cochain on A . Thus for a function from singular n -simplices in X to G , the image of this function under i^* is obtained by restricting the domain of the function to singular n -simplices in A . Every function from singular n -simplices in A to G can be extended to be defined on all singular n -simplices in X , for example by assigning the value 0 to all singular n -simplices not in A , so i^* is surjective. The kernel of i^* consists of cochains taking the value 0 on singular n -simplices in A . Such cochains are the same as homomorphisms $C_n(X, A) = C_n(X)/C_n(A) \rightarrow G$, so the kernel of i^* is exactly $C^n(X, A; G) = \text{Hom}(C_n(X, A), G)$, giving the desired exactness. Notice that we can view $C^n(X, A; G)$ as the functions from singular n -simplices in X to G that vanish on simplices in A , since the basis for $C_n(X)$ consisting of singular n -simplices in X is the disjoint union of the simplices with image contained in A and the simplices with image not contained in A .

Relative coboundary maps $\delta: C^n(X, A; G) \rightarrow C^{n+1}(X, A; G)$ are obtained as restrictions of the absolute δ 's, so relative cohomology groups $H^n(X, A; G)$ are defined. The

fact that the relative cochain group is a subgroup of the absolute cochains, namely the cochains vanishing on chains in A , means that relative cohomology is conceptually a little simpler than relative homology.

The maps i^* and j^* commute with δ since i and j commute with ∂ , so the preceding displayed short exact sequence of cochain groups is part of a short exact sequence of cochain complexes, giving rise to an associated long exact sequence of cohomology groups

$$\cdots \rightarrow H^n(X, A; G) \xrightarrow{j^*} H^n(X; G) \xrightarrow{i^*} H^n(A; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \rightarrow \cdots$$

By similar reasoning one obtains a long exact sequence of reduced cohomology groups for a pair (X, A) with A nonempty, where $\tilde{H}^n(X, A; G) = H^n(X, A; G)$ for all n , as in homology. Taking A to be a point x_0 , this exact sequence gives an identification of $\tilde{H}^n(X; G)$ with $H^n(X, x_0; G)$.

More generally there is a long exact sequence for a triple (X, A, B) coming from the short exact sequences

$$0 \leftarrow C^n(A, B; G) \xleftarrow{i^*} C^n(X, B; G) \xleftarrow{j^*} C^n(X, A; G) \leftarrow 0$$

The long exact sequence of reduced cohomology can be regarded as the special case that B is a point.

As one would expect, there is a duality relationship between the connecting homomorphisms $\delta: H^n(A; G) \rightarrow H^{n+1}(X, A; G)$ and $\partial: H_{n+1}(X, A) \rightarrow H_n(A)$. This takes the form of the commutative diagram shown at the right. To verify commutativity, recall how the two connecting homomorphisms are defined, via the diagrams

$$\begin{array}{ccc} H^n(A; G) & \xrightarrow{\delta} & H^{n+1}(X, A; G) \\ \downarrow h & & \downarrow h \\ \text{Hom}(H_n(A), G) & \xrightarrow{\partial^*} & \text{Hom}(H_{n+1}(X, A), G) \end{array}$$

$$\begin{array}{ccccc} C^{n+1}(X; G) & \xleftarrow{\quad} & C^{n+1}(X, A; G) & & C_{n+1}(X; G) & \xrightarrow{\quad} & C_{n+1}(X, A; G) \\ & \nearrow \text{dashed} & \uparrow & & \nwarrow \text{dashed} & & \\ C^n(A; G) & \xleftarrow{\quad} & C^n(X; G) & & C_n(A; G) & \xrightarrow{\quad} & C_n(X; G) \end{array}$$

The connecting homomorphisms are represented by the dashed arrows, which are well-defined only when the chain and cochain groups are replaced by homology and cohomology groups. To show that $h\delta = \partial^*h$, start with an element $\alpha \in H^n(A; G)$ represented by a cocycle $\varphi \in C^n(A; G)$. To compute $\delta(\alpha)$ we first extend φ to a cochain $\bar{\varphi} \in C^n(X; G)$, say by letting it take the value 0 on singular simplices not in A . Then we compose $\bar{\varphi}$ with $\partial: C_{n+1}(X) \rightarrow C_n(X)$ to get a cochain $\bar{\varphi}\partial \in C^{n+1}(X; G)$, which actually lies in $C^{n+1}(X, A; G)$ since the original φ was a cocycle in A . This cochain $\bar{\varphi}\partial \in C^{n+1}(X, A; G)$ represents $\delta(\alpha)$ in $H^{n+1}(X, A; G)$. Now we apply the map h , which simply restricts the domain of $\bar{\varphi}\partial$ to relative cycles in $C_{n+1}(X, A)$, that is, $(n+1)$ -chains in X whose boundary lies in A . On such chains we have $\bar{\varphi}\partial = \varphi\partial$ since the extension of φ to $\bar{\varphi}$ is irrelevant. The net result of all this is that $h\delta(\alpha)$

is represented by $\varphi\partial$. Let us compare this with $\partial^*h(\alpha)$. Applying h to φ restricts its domain to cycles in A . Then applying ∂^* composes with the map which sends a relative $(n+1)$ -cycle in X to its boundary in A . Thus $\partial^*h(\alpha)$ is represented by $\varphi\partial$ just as $h\delta(\alpha)$ was, and so the square commutes.

Induced Homomorphisms. Dual to the chain maps $f_\# : C_n(X) \rightarrow C_n(Y)$ induced by $f : X \rightarrow Y$ are the cochain maps $f^\# : C^n(Y; G) \rightarrow C^n(X; G)$. The relation $f_\#\partial = \partial f_\#$ dualizes to $\delta f^\# = f^\# \delta$, so $f^\#$ induces homomorphisms $f^* : H^n(Y; G) \rightarrow H^n(X; G)$. In the relative case a map $f : (X, A) \rightarrow (Y, B)$ induces $f^* : H^n(Y, B; G) \rightarrow H^n(X, A; G)$ by the same reasoning, and in fact f induces a map between short exact sequences of cochain complexes, hence a map between long exact sequences of cohomology groups, with commuting squares. The properties $(fg)^\# = g^\# f^\#$ and $\mathbb{1}^\# = \mathbb{1}$ imply $(fg)^* = g^* f^*$ and $\mathbb{1}^* = \mathbb{1}$, so $X \mapsto H^n(X; G)$ and $(X, A) \mapsto H^n(X, A; G)$ are contravariant functors, the ‘contra’ indicating that induced maps go in the reverse direction.

The algebraic universal coefficient theorem applies also to relative cohomology since the relative chain groups $C_n(X, A)$ are free, and there is a naturality statement: A map $f : (X, A) \rightarrow (Y, B)$ induces a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ext}(H_{n-1}(X, A), G) & \longrightarrow & H^n(X, A; G) & \xrightarrow{h} & \text{Hom}(H_n(X, A), G) \longrightarrow 0 \\ & & \uparrow (f_\#)^* & & \uparrow f^* & & \uparrow (f_\#)^* \\ 0 & \longrightarrow & \text{Ext}(H_{n-1}(Y, B), G) & \longrightarrow & H^n(Y, B; G) & \xrightarrow{h} & \text{Hom}(H_n(Y, B), G) \longrightarrow 0 \end{array}$$

This follows from the naturality of the algebraic universal coefficient sequences since the vertical maps are induced by the chain maps $f_\# : C_n(X, A) \rightarrow C_n(Y, B)$. When the subspaces A and B are empty we obtain the absolute forms of these results.

Homotopy Invariance. The statement is that if $f \simeq g : (X, A) \rightarrow (Y, B)$, then $f^* = g^* : H^n(Y, B) \rightarrow H^n(X, A)$. This is proved by direct dualization of the proof for homology. From the proof of Theorem 2.10 we have a chain homotopy P satisfying $g_\# - f_\# = \partial P + P\partial$. This relation dualizes to $g^\# - f^\# = P^* \delta + \delta P^*$, so P^* is a chain homotopy between the maps $f^\#, g^\# : C^n(Y; G) \rightarrow C^n(X; G)$. This restricts also to a chain homotopy between $f^\#$ and $g^\#$ on relative cochains, the cochains vanishing on singular simplices in the subspaces B and A . Since $f^\#$ and $g^\#$ are chain homotopic, they induce the same homomorphism $f^* = g^*$ on cohomology.

Excision. For cohomology this says that for subspaces $Z \subset A \subset X$ with the closure of Z contained in the interior of A , the inclusion $i : (X - Z, A - Z) \hookrightarrow (X, A)$ induces isomorphisms $i^* : H^n(X, A; G) \rightarrow H^n(X - Z, A - Z; G)$ for all n . This follows from the corresponding result for homology by the naturality of the universal coefficient theorem and the five-lemma. Alternatively, if one wishes to avoid appealing to the universal coefficient theorem, the proof of excision for homology dualizes easily to cohomology by the following argument. In the proof for homology there were chain maps $\iota : C_n(A + B) \rightarrow C_n(X)$ and $\rho : C_n(X) \rightarrow C_n(A + B)$ such that $\rho\iota = \mathbb{1}$ and $\mathbb{1} - \iota\rho = \partial D + D\partial$ for a chain homotopy D . Dualizing by taking $\text{Hom}(-, G)$, we have maps

ρ^* and ι^* between $C^n(A+B;G)$ and $C^n(X;G)$, and these induce isomorphisms on cohomology since $\iota^*\rho^* = \mathbb{1}$ and $\mathbb{1} - \rho^*\iota^* = D^*\delta + \delta D^*$. By the five-lemma, the maps $C^n(X, A; G) \rightarrow C^n(A+B, A; G)$ also induce isomorphisms on cohomology. There is an obvious identification of $C^n(A+B, A; G)$ with $C^n(B, A \cap B; G)$, so we get isomorphisms $H^n(X, A; G) \approx H^n(B, A \cap B; G)$ induced by the inclusion $(B, A \cap B) \hookrightarrow (X, A)$.

Axioms for Cohomology. These are exactly dual to the axioms for homology. Restricting attention to CW complexes again, a (reduced) **cohomology theory** is a sequence of contravariant functors $\tilde{h}^n$ from CW complexes to abelian groups, together with natural coboundary homomorphisms $\delta: \tilde{h}^n(A) \rightarrow \tilde{h}^{n+1}(X/A)$ for CW pairs (X, A) , satisfying the following axioms:

- (1) If $f \simeq g: X \rightarrow Y$, then $f^* = g^*: \tilde{h}^n(Y) \rightarrow \tilde{h}^n(X)$.
- (2) For each CW pair (X, A) there is a long exact sequence

$$\dots \xrightarrow{\delta} \tilde{h}^n(X/A) \xrightarrow{q^*} \tilde{h}^n(X) \xrightarrow{i^*} \tilde{h}^n(A) \xrightarrow{\delta} \tilde{h}^{n+1}(X/A) \xrightarrow{q^*} \dots$$

where i is the inclusion and q is the quotient map.

- (3) For a wedge sum $X = \bigvee_{\alpha} X_{\alpha}$ with inclusions $i_{\alpha}: X_{\alpha} \hookrightarrow X$, the product map $\prod_{\alpha} i_{\alpha}^*: \tilde{h}^n(X) \rightarrow \prod_{\alpha} \tilde{h}^n(X_{\alpha})$ is an isomorphism for each n .

We have already seen that the first axiom holds for singular cohomology. The second axiom follows from excision in the same way as for homology, via isomorphisms $\tilde{H}^n(X/A; G) \approx H^n(X, A; G)$. Note that the third axiom involves direct product, rather than the direct sum appearing in the homology version. This is because of the natural isomorphism $\text{Hom}(\bigoplus_{\alpha} A_{\alpha}, G) \approx \prod_{\alpha} \text{Hom}(A_{\alpha}, G)$, which implies that the cochain complex of a disjoint union $\coprod_{\alpha} X_{\alpha}$ is the direct product of the cochain complexes of the individual X_{α} 's, and this direct product splitting passes through to cohomology groups. The same argument applies in the relative case, so we get isomorphisms $H^n(\coprod_{\alpha} X_{\alpha}, \coprod_{\alpha} A_{\alpha}; G) \approx \prod_{\alpha} H^n(X_{\alpha}, A_{\alpha}; G)$. The third axiom is obtained by taking the A_{α} 's to be basepoints x_{α} and passing to the quotient $\coprod_{\alpha} X_{\alpha} / \coprod_{\alpha} x_{\alpha} = \bigvee_{\alpha} X_{\alpha}$.

The relation between reduced and unreduced cohomology theories is the same as for homology, as described in §2.3.

Simplicial Cohomology. If X is a Δ -complex and $A \subset X$ is a subcomplex, then the simplicial chain groups $\Delta_n(X, A)$ dualize to simplicial cochain groups $\Delta^n(X, A; G) = \text{Hom}(\Delta_n(X, A), G)$, and the resulting cohomology groups are by definition the simplicial cohomology groups $H_{\Delta}^n(X, A; G)$. Since the inclusions $\Delta_n(X, A) \subset C_n(X, A)$ induce isomorphisms $H_{\Delta}^n(X, A) \approx H_n(X, A)$, Corollary 3.4 implies that the dual maps $C^n(X, A; G) \rightarrow \Delta^n(X, A; G)$ also induce isomorphisms $H^n(X, A; G) \approx H_{\Delta}^n(X, A; G)$.

Cellular Cohomology. For a CW complex X this is defined via the cellular cochain complex formed by the horizontal sequence in the following diagram, where coefficients in a given group G are understood, and the cellular coboundary maps d_n are

the compositions $\delta_n j_n$, making the triangles commute. Note that $d_n d_{n-1} = 0$ since $j_n \delta_{n-1} = 0$.

$$\begin{array}{ccccccc}
 & & & & 0 & & \\
 & & & \nearrow & & \searrow & \\
 & & H^{n-1}(X^{n-1}) & & & & \\
 j_{n-1} \nearrow & & & \searrow \delta_{n-1} & & & \\
 \cdots \longrightarrow & H^{n-1}(X^{n-1}, X^{n-2}) & \xrightarrow{d_{n-1}} & H^n(X^n, X^{n-1}) & \xrightarrow{d_n} & H^{n+1}(X^{n+1}, X^n) & \longrightarrow \cdots \\
 & & & \searrow j_n & \nearrow \delta_n & & \\
 & & & H^n(X^n) & & & \\
 & & \nearrow & & \searrow & & \\
 & H^n(X) \approx H^n(X^{n+1}) & & & & 0 & \\
 0 \nearrow & & & & & &
 \end{array}$$

Theorem 3.5. $H^n(X; G) \approx \text{Ker } d_n / \text{Im } d_{n-1}$. Furthermore, the cellular cochain complex $\{H^n(X^n, X^{n-1}; G), d_n\}$ is isomorphic to the dual of the cellular chain complex, obtained by applying $\text{Hom}(-, G)$.

Proof: The universal coefficient theorem implies that $H^k(X^n, X^{n-1}; G) = 0$ for $k \neq n$. The long exact sequence of the pair (X^n, X^{n-1}) then gives isomorphisms $H^k(X^n; G) \approx H^k(X^{n-1}; G)$ for $k \neq n, n-1$. Hence by induction on n we obtain $H^k(X^n; G) = 0$ if $k > n$. Thus the diagonal sequences in the preceding diagram are exact. The universal coefficient theorem also gives $H^k(X, X^{n+1}; G) = 0$ for $k \leq n+1$, so $H^n(X; G) \approx H^n(X^{n+1}; G)$. The diagram then yields isomorphisms

$$H^n(X; G) \approx H^n(X^{n+1}; G) \approx \text{Ker } \delta_n \approx \text{Ker } d_n / \text{Im } \delta_{n-1} \approx \text{Ker } d_n / \text{Im } d_{n-1}$$

For the second statement in the theorem we have the diagram

$$\begin{array}{ccccc}
 H^k(X^k, X^{k-1}; G) & \longrightarrow & H^k(X^k; G) & \xrightarrow{\delta} & H^{k+1}(X^{k+1}, X^k; G) \\
 \downarrow h & & \downarrow h & & \downarrow h \\
 \text{Hom}(H_k(X^k, X^{k-1}), G) & \longrightarrow & \text{Hom}(H_k(X^k), G) & \xrightarrow{\partial^*} & \text{Hom}(H_{k+1}(X^{k+1}, X^k), G)
 \end{array}$$

The cellular coboundary map is the composition across the top, and we want to see that this is the same as the composition across the bottom. The first and third vertical maps are isomorphisms by the universal coefficient theorem, so it suffices to show the diagram commutes. The first square commutes by naturality of h , and commutativity of the second square was shown in the discussion of the long exact sequence of cohomology groups of a pair (X, A) . $\square$

Mayer-Vietoris Sequences. In the absolute case these take the form

$$\cdots \rightarrow H^n(X; G) \xrightarrow{\Psi} H^n(A; G) \oplus H^n(B; G) \xrightarrow{\Phi} H^n(A \cap B; G) \rightarrow H^{n+1}(X; G) \rightarrow \cdots$$

where X is the union of the interiors of A and B . This is the long exact sequence associated to the short exact sequence of cochain complexes

$$0 \rightarrow C^n(A+B, G) \xrightarrow{\Psi} C^n(A; G) \oplus C^n(B; G) \xrightarrow{\Phi} C^n(A \cap B; G) \rightarrow 0$$

Here $C^n(A+B;G)$ is the dual of the subgroup $C_n(A+B) \subset C_n(X)$ consisting of sums of singular n -simplices lying in A or in B . The inclusion $C_n(A+B) \subset C_n(X)$ is a chain homotopy equivalence by Proposition 2.21, so the dual restriction map $C^n(X;G) \rightarrow C^n(A+B;G)$ is also a chain homotopy equivalence, hence induces an isomorphism on cohomology as shown in the discussion of excision a couple pages back. The map ψ has coordinates the two restrictions to A and B , and φ takes the difference of the restrictions to $A \cap B$, so it is obvious that φ is onto with kernel the image of ψ .

There is a relative Mayer-Vietoris sequence

$$\cdots \rightarrow H^n(X, Y; G) \rightarrow H^n(A, C; G) \oplus H^n(B, D; G) \rightarrow H^n(A \cap B, C \cap D; G) \rightarrow \cdots$$

for a pair $(X, Y) = (A \cup B, C \cup D)$ with $C \subset A$ and $D \subset B$ such that X is the union of the interiors of A and B while Y is the union of the interiors of C and D . To derive this, consider first the map of short exact sequences of cochain complexes

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^n(X, Y; G) & \longrightarrow & C^n(X; G) & \longrightarrow & C^n(Y; G) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C^n(A+B, C+D; G) & \longrightarrow & C^n(A+B; G) & \longrightarrow & C^n(C+D; G) \longrightarrow 0 \end{array}$$

Here $C^n(A+B, C+D; G)$ is defined as the kernel of $C^n(A+B; G) \rightarrow C^n(C+D; G)$, the restriction map, so the second sequence is exact. The vertical maps are restrictions. The second and third of these induce isomorphisms on cohomology, as we have seen, so by the five-lemma the first vertical map also induces isomorphisms on cohomology. The relative Mayer-Vietoris sequence is then the long exact sequence associated to the short exact sequence of cochain complexes

$$0 \rightarrow C^n(A+B, C+D; G) \xrightarrow{\psi} C^n(A, C; G) \oplus C^n(B, D; G) \xrightarrow{\varphi} C^n(A \cap B, C \cap D; G) \rightarrow 0$$

This is exact since it is the dual of the short exact sequence

$$0 \rightarrow C_n(A \cap B, C \cap D) \rightarrow C_n(A, C) \oplus C_n(B, D) \rightarrow C_n(A+B, C+D) \rightarrow 0$$

constructed in §2.2, which splits since $C_n(A+B, C+D)$ is free with basis the singular n -simplices in A or in B that do not lie in C or in D .

Exercises

1. Show that $\text{Ext}(H, G)$ is a contravariant functor of H for fixed G , and a covariant functor of G for fixed H .
2. Show that the maps $G \xrightarrow{n} G$ and $H \xrightarrow{n} H$ multiplying each element by the integer n induce multiplication by n in $\text{Ext}(H, G)$.
3. Regarding $\mathbb{Z}_2$ as a module over the ring $\mathbb{Z}_4$, construct a resolution of $\mathbb{Z}_2$ by free modules over $\mathbb{Z}_4$ and use this to show that $\text{Ext}_{\mathbb{Z}_4}^n(\mathbb{Z}_2, \mathbb{Z}_2)$ is nonzero for all n .